

Solution Architecture

Date	29 September 2026
Team ID	6ab2a427dbd693aadc5a7d4b
Project Name	Student Enrollment & Course Management System (SECMS)
Maximum Marks	2 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	UI for Students Admin Faculty - enroll view courses pay fees verify - responsive entry point	React.js HTML5 CSS3 Bootstrap 5 Axios
API Gateway / Load Balancer	Single entry point routes requests load balancing during peak rate limiting JWT check	Node.js Express.js Nginx AWS API Gateway
Auth Service	Manages auth - registration OTP JWT Google SSO role check	JWT bcrypt OAuth2 Google Auth
Core Logic Service	Heart - enrollment logic prerequisite check seat locking auto-allotment fee logic approval	Node.js Express Sequelize ORM
External API Integrations	Third party payment communication storage	Razorpay SendGrid Twilio AWS S3
Database / Data Layer	Persistent storage ACID to avoid double booking	MySQL MongoDB AWS S3 RDS